Clinical Text or Corpus Structure? Medical Specialty Classification with Sparse Models and Clinical BERT

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1 Introduction

Classifying clinical records into medical specialties is challenging due to heterogeneous documentation and overlap between clinical domains. Clinical narratives often contain information relevant to multiple specialties, making their automatic categorisation ambiguous. This project explores natural language processing approaches to specialty classification using the textual information available in clinical records. Specifically, it addresses the following research question: *How effectively can natural language processing models classify clinical records into medical specialties using clinical text alone, and how much of the attainable performance comes instead from structural properties of the corpus?*

The task involves 12 medical specialties. Different classification models are evaluated using overall and per-class performance metrics.

2 Data

2.1 Data Analysis

The training dataset comprises 2,669 clinical records across 12 medical specialties. Each record contains a specialty label and four input fields: description, sample name, transcription, and keywords. The class distribution is highly imbalanced: Surgery accounts for 893 records (33.5%), whereas Dermatology contains only 25 (0.9%). The class distribution is shown in Figure 2 in Appendix A. This imbalance motivates the use of per-class metrics alongside overall performance measures.

The source CSV contained seven records with missing fields, six lines with additional empty fields, and three transcriptions split across consecutive lines. We filled missing fields with empty strings, removed empty trailing fields, and rejoined the fragmented transcriptions. No records were discarded during this structural correction. The resulting dataset contains one empty description, seven empty sample names, 31 empty transcriptions, and 450 records without keywords.

Inspection also revealed that the first keyword matched the assigned medical specialty in all 2,219 records with non-empty keywords. We therefore removed the first keyword from each record while retaining the remaining keywords, allowing us to evaluate the models without explicit specialty identifiers. In the test file, 320 of the 398 records also start with a specialty name.

Finally, 1,528 records (57.2%) share an identical non-empty transcription with at least one record assigned to a different specialty. For example, two records describing the same lower-extremity venous Doppler examination have identical input fields but are labelled Cardiovascular-Pulmonary and Radiology. These repeated texts with different labels introduce ambiguity into the classification task. Moreover, 239 of the 398 test transcriptions also occur in the training data. Their treatment is therefore considered in the experimental design, including configurations that consolidate repeated transcriptions and retain the most frequent specialty label within each group.

2.2 Data Augmentation

We used data augmentation only for Model 2, which was designed to improve upon Model 1 by addressing class imbalance. Following easy data augmentation (EDA) [Wei and Zou, 2019], we expanded the classes with fewer than 100 training records by adding 3 augmented copies of each record, using two strategies: random swapping (3 word pairs) and swapping combined with deletion (each word deleted with probability 0.1, then 5% of positions swapped). This increases the training split from 2,135 to 2,657 records; each strategy is run with 3 augmentation seeds and the mean is reported. Synonym replacement was not used because general-purpose thesauri do not adequately cover clinical vocabulary. Augmentation is applied to the training split only: augmenting before the split places near-identical copies of a record on both sides, which in our tests inflated accuracy from 77.3% to 84.1% and the minimum per-class F1 from 0.33 to 0.69.

The objective was to prevent the classifier from favouring larger classes, such as Surgery, and to improve performance on smaller classes, such as Dermatology and Ophthalmology. However, these approaches can produce grammatically incorrect or less realistic medical sentences, introducing noise into the dataset, an issue that could be particularly concerning in real clinical settings. Whether augmentation actually helps is examined in Section 6.

3 Models

All text models read the four fields concatenated, with the first keyword removed, so that they learn

from clinical content. We also added two rules on top, as they were clearly shown in the data:

Keyword rule (R1): if the first keyword names a specialty, it is returned.

Duplicate-note rule (R2): if the transcription of a record occurs in the training data, the labels of its training copies are excluded and the highest-scoring remaining label is returned. This rule uses training data only.

For our baseline we used Multinomial Naive Bayes (MNB) on term frequency–inverse document frequency (TF-IDF) features of the description, lowercased and Snowball-stemmed (the baseline defined in the project statement).

Model 1: TF-IDF + linear support vector machine (SVM).

Lower-cased word unigrams, sublinear term frequency ($1 + \log \mathrm{tf}$) and a minimum document frequency of 2. Stop words are kept, since TF-IDF already assigns them low weights. A SVM with balanced class weights (each class weighted inversely to its frequency) addresses the imbalance.

Model 2: n-grams + logistic regression (LR) + augmentation.

Word uni/bigrams (to capture terms such as "blood pressure" or "no fracture"), a class-balanced LR, and the augmentation of Section 2.2. We also evaluate collapsing repeated transcriptions into one record with the most frequent specialty (mode de-duplication).

Model 3: fine-tuned Clinical BERT.

We fine-tuned Bio_ClinicalBERT, a BERT model [Devlin et al., 2019] pre-trained on biomedical literature and on clinical notes [Alsentzer et al., 2019], with a linear classification layer on the `[CLS]` representation. All weights are fine-tuned. No lowercasing, stemming or stop-word removal is applied, as the model uses its own WordPiece tokenizer and was pre-trained on raw text. Records have a median of 746 tokens and 72% exceed BERT's limit of 512 tokens, so the input is ordered from most important to least *sample name [SEP] description [SEP] keywords [SEP] transcription*: truncation then removes only the end of the transcription. Class imbalance is handled with a class-weighted cross-entropy loss, making sure rarer classes are given more importance.

The submitted system is Model 3 with the two rules stated above.

4 Experimental Setup

The official test set was used only to produce `results.txt`. All results come from our own test set: a stratified 80/20 split of the training data (2,135 / 534 records, `random_state=42`), identical for all models. The split is random rather than grouped by transcription, so that, as in the official test set, 240 of the 534 test records have a copy in the training split. Metrics are accuracy, macro-F1 and the minimum per-class F1. Each model is reported (i) alone, (ii) with R2, (iii) with R1 and R2 (*full*), and (iv) on the 94 test records without keywords, which are the only records decided by the text model in the full system (*no-KW*).

Model	alone acc	+R2 acc	full (R1+R2) acc	mF1	minF1	no-KW acc
Baseline (MNB)	39.3	54.5	88.2	84.5	22.2	33.0
M1 TF-IDF + SVM	44.2	76.2	94.6	93.1	79.2	69.1
M2 n-grams + LR	40.1	77.3	94.8	93.3	79.6	70.2
+ random swap	38.0	77.4	94.8	93.3	79.6	70.2
+ swap + deletion	37.7	77.3	94.8	93.4	79.4	70.2
+ mode de-dup.	38.6	76.6	94.8	93.3	80.4	70.2
M3 Clinical BERT	54.9	75.5	95.1	92.7	80.0	72.3
mean of 3 seeds	55.2	76.5	95.8	93.6	–	75.9
± std	0.4	0.9	0.7	1.0	–	3.7
3-seed ensemble	–	–	95.9	94.4	85.1	76.6

Table 1: Results (%) on our test set (534 records; no-KW: the 94 records without keywords). M3 rows without annotation use seed 42, the submitted model.

Class	P	R	F1	n	Class	P	R	F1	n
Cardio	98.2	87.1	92.3	62	Ortho	98.4	100	99.2	60
Derm	80.0	80.0	80.0	5	ObGyn	96.2	92.6	94.3	27
Gastro	92.5	94.9	93.7	39	Ophth	93.8	100	96.8	15
GenMed	73.6	92.9	82.1	42	Psych	87.5	87.5	87.5	8
Neuro	94.4	91.9	93.2	37	Radio	97.7	97.7	97.7	44
NSurg	100	93.8	96.8	16	Surgery	100	97.8	98.9	179

Table 2: Per-class precision, recall and F1 (%) of the submitted system (Model 3 + R1 + R2, seed 42) on our test set.

4.1 Parameters/Hyperparameters

Model 1: regularisation constant $C = 0.3$ (inverse strength of the penalty on large weights). **Model 2:** $C = 10$, L2 penalty, 3,000 iterations. **Model 3:** 4 epochs, learning rate $2 \cdot 10^{-5}$ and weight decay 0.01, linear warm-up over the first 10% of steps followed by linear decay, batch size 8 with 2 gradient-accumulation steps (effective batch 16), gradient clipping at norm 1.0, bf16 mixed precision, maximum length 512 tokens. Model 3 is trained with seeds 42, 7 and 123; seed 42 produces `results.txt`.

5 Results

Table 1 summarises the experiments. The best system is Model 3 with both rules: 95.1% accuracy, 92.7% macro-F1 and a minimum per-class F1 of 80.0% (seed 42), and $95.8 \pm 0.7\%$ accuracy over three seeds. Table 2 gives its per-class scores. All 26 errors of this system occur on records without keywords: R1 labelled the other 440 test records correctly. Figure 1 (next page) is the confusion matrix of Clinical BERT with R2, which shows the behaviour of the text model on all 534 records.

6 Discussion

Corpus structure dominates. For Models 1–3, R2 adds 21 to 37 points of accuracy and R1 a further 18 to 20 (Table 1). Without R2, the models memorise

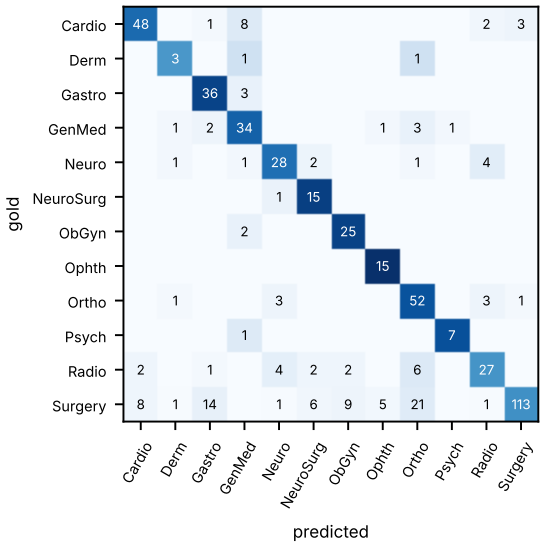


Figure 1: Confusion matrix of Clinical BERT with R2 (without the keyword rule) on our test set: counts, colour normalised per gold class.

the label of a training note and are wrong on its test copy; with R2, this becomes an advantage. With both rules, Models 1–3 differ by at most 0.5 points (seed 42), that is, 3 of 534 records.

Where the text model fails. Figure 1 shows that 66 of the 179 Surgery records are predicted as an organ specialty (21 Orthopedic, 14 Gastroenterology, 9 Obstetrics-Gynecology, 8 Cardiovascular-Pulmonary), and that Radiology records are predicted as Orthopedic (6) and Neurology (4). These are the specialty pairs under which the same notes are listed: the model identifies the organ, whereas the gold label records the type of act (operation or imaging).

Clinical text alone. The text models are compared on the records they actually decide (no-KW). Clinical BERT reaches $75.9 \pm 3.7\%$ against 69.1% for the SVM and 70.2% for the LR; each of the three seeds exceeds both sparse models, although with 94 records one record is one point. Without the rules, BERT is 10.7 points above the SVM (54.9 vs. 44.2%). Pre-training on clinical text therefore helps, but the text models remain about 20 points below the full system.

Augmentation and de-duplication. Neither augmentation changes the no-KW accuracy (70.2% in all configurations), and both lower the accuracy without rules (40.1 $\rightarrow$ 38.0 and 37.7%). The remaining errors stem from records whose correct label is ambiguous, not from a lack of minority-class examples, and the class weights already compensate for the imbalance. Mode de-duplication breaks almost every tie alphabetically: of the repeated transcriptions, none keeps the label Surgery, and the minimum per-class F1 without rules drops from 18.2 to 11.1%.

Misclassified examples (submitted system, own test set): (1) “Initial clinic visit for foreign body in left eye.” (sample name “Gen Med Consult”): gold General Medicine, predicted Ophthalmology. The content is ophthalmic; the gold label reflects the consultation setting, not the organ. (2) “Left heart catheterization, coronary angiography, and left ventriculogram. No angiographic evidence of coronary artery disease”: gold Surgery, predicted Cardiovascular-Pulmonary. A cardiology procedure; both labels are defensible. (3) “Obstructive sleep apnea syndrome. Loud snoring. Schedule an overnight sleep study” (“Pulmonary Consultation”): gold Cardiovascular-Pulmonary, predicted General Medicine. (4) “Preeclampsia, status post delivery with Cesarean section with uncontrolled blood pressure”: gold Obstetrics-Gynecology, predicted General Medicine. In (3) and (4), and in several emergency-room notes (“Difficulty Breathing – ER Visit”, gold Cardiovascular-Pulmonary), the note is an outpatient consultation or admission about blood pressure or breathing, a layout that is frequent in General Medicine (212 records, 90 of them without keywords). The model relies on the document type more than on the organ named in the sample name; General Medicine has the lowest precision (73.6%, Table 2). (1) and (2) are cases where the gold label is one of several reasonable answers.

Answer to the research question. Using clinical text alone, the best model (Clinical BERT) classifies 55% of our test records correctly when notes are shared with the training set, and 76% of the records without keywords once shared notes are handled. Most of the 95.8% of the final system comes from two corpus properties, not from the clinical text. Within the text models, pre-training on clinical notes yields a consistent but modest gain, and augmentation yields none.

7 Future Work

A multi-label formulation, in which each note is assigned the set of specialties it is listed under rather than a single one, would directly model the fact that the same transcription appears under several specialties, and would evaluate text models without penalising predictions that match another valid listing of the same note. For the text model, we would read the 72% of records longer than 512 tokens in chunks or with a long-context clinical model, train on soft labels reflecting each note’s label distribution, and evaluate with grouped cross-validation over more seeds.

Bibliography

Use of LLMs. Claude was used to write and debug the code, draft and revise the text of this paper. All reported numbers were produced by the submitted notebooks; the group reviewed the code, the outputs and the text.

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Appendix A: Extra Figures and Tables

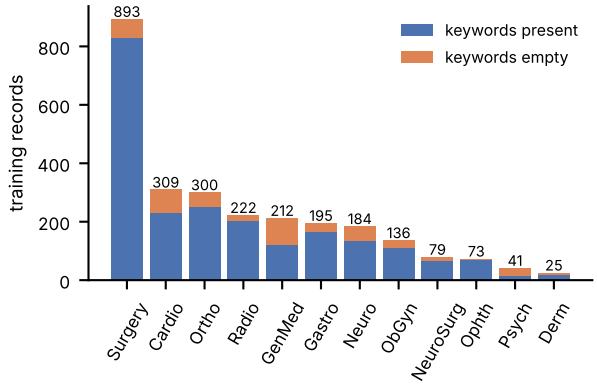


Figure 2: Training records per specialty, split by whether the keywords field is empty.